

ARUNKUMAR S

AI/ML Engineer — Computer Vision & Generative AI

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SUMMARY

AI/ML Engineer with 2+ years building production computer vision and generative AI systems for security-sensitive, air-gapped environments. Delivered a fully offline enterprise RAG platform and TTS engine for a defense-sector client, a YOLOv8 crash-detection model running at ~87% mAP for a leading commercial vehicle OEM, and LoRA-tuned LLMs improving domain accuracy by ~12%. Comfortable owning a system end-to-end model training, API design, and deployment with a consistent record of shipping measurable accuracy and efficiency gains rather than proofs of concept.

EXPERIENCE

Jr. Software Engineer

GGG Information Services Pvt. Ltd

06/2024 – Present

Bangalore, India

- Defense-Sector Client - Offline RAG Platform:** Designed and built a fully offline Retrieval-Augmented Generation platform (Ollama, Qdrant, Podman, WSL2) for a client with strict data-residency requirements, covering PDF ingestion, semantic chunking, embedding generation, and context-aware retrieval.
- Defense-Sector Client - Offline TTS Engine:** Built a fully offline text-to-speech engine with word-level synchronized highlighting and an intelligent text-normalization layer for engineering abbreviations, units, and symbols; a custom caching layer cut repeated-inference latency and kept playback responsive.
- Commercial Vehicle OEM - Crash Impact Detection:** Trained and deployed a YOLOv8/OpenCV model for real-time vehicle damage assessment, reaching ~87% mAP across three severity classes and cutting manual inspection effort by ~50%.
- Automotive OEM - 3D-to-2D Documentation Automation:** Built a Blender + Python pipeline that programmatically explodes CAD assemblies and generates callouts, converting 3D models into structured SVG line-art and reducing manual technical-illustration time by over 70%.
- LLM Fine-Tuning:** Applied LoRA/PEFT fine-tuning on Qwen for domain-specific document understanding, lifting response accuracy ~12% over the base model on automotive service-query benchmarks.
- Agentic Workflows:** Orchestrated LangChain and LangGraph agents across 4+ tools — ingestion, retrieval, reasoning, generation — into autonomous document Q&A pipelines for enterprise knowledge bases, using LangSmith to trace, evaluate, and debug multi-step agent runs.
- API & Platform:** Built and maintain 3 production Flask/FastAPI services, including the crash-detection and RAG Q&A systems above, with centralized error handling and structured JSON I/O for reliable inference at scale.

PROJECTS

RAG-Powered Gen-AI Chatbot with Voice Interface — *Self-Directed, Personal Time*

- Designed and built a full-stack RAG chatbot end-to-end in personal time — independent of client engagements — using Flask, FAISS, LangChain, and AWS Bedrock, supporting PDF upload, automatic text extraction, and vector-based retrieval, and reaching ~83% retrieval accuracy with measurably reduced hallucinations.
- Extended the chatbot with a full voice interface using AWS Polly (TTS) and Transcribe (STT), adding speech-based interaction on top of the retrieval pipeline.

Technologies: Python • AWS Bedrock • LangChain • FAISS • Flask • AWS Polly • AWS Transcribe

EDUCATION

Bachelor of Technology, Information Technology

SRM Easwari Engineering College

2020 – 2024

Chennai, India | CGPA: 8.54/10

TECHNICAL SKILLS

Backend & APIs: Python, Flask, FastAPI, REST APIs, SQL, NumPy, Pandas, scikit-learn

Generative AI: RAG, Agentic AI, LangChain, LangGraph, LangSmith, FAISS, Qdrant, Ollama, AWS Bedrock, LoRA/PEFT, NLP, Prompt Engineering, Vector Databases, Hugging Face

AI-Native / Agentic Development: GitHub Copilot, Cursor, Claude Code

Computer Vision: PyTorch, TensorFlow, YOLO (Darknet, Ultralytics), OpenCV, CNN, LSTM, Transformers, MediaPipe, OpenPose, OCR (PaddleOCR, Tesseract)

Automation: Blender Python API, SVG Generation, CAD Automation

Cloud (AWS): Bedrock, Polly, Transcribe, API Gateway, Textract

DevOps & Deployment: Docker, Podman, WSL2, Git, GitHub

Model Optimization: ONNX, ONNX Runtime, Model Quantization (INT8/GGUF), llama.cpp, On-Premise LLM Inference